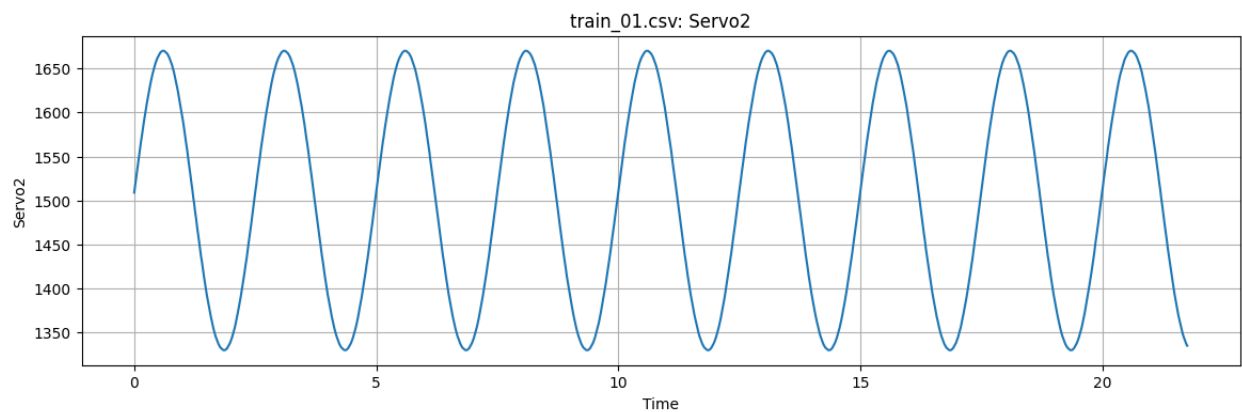
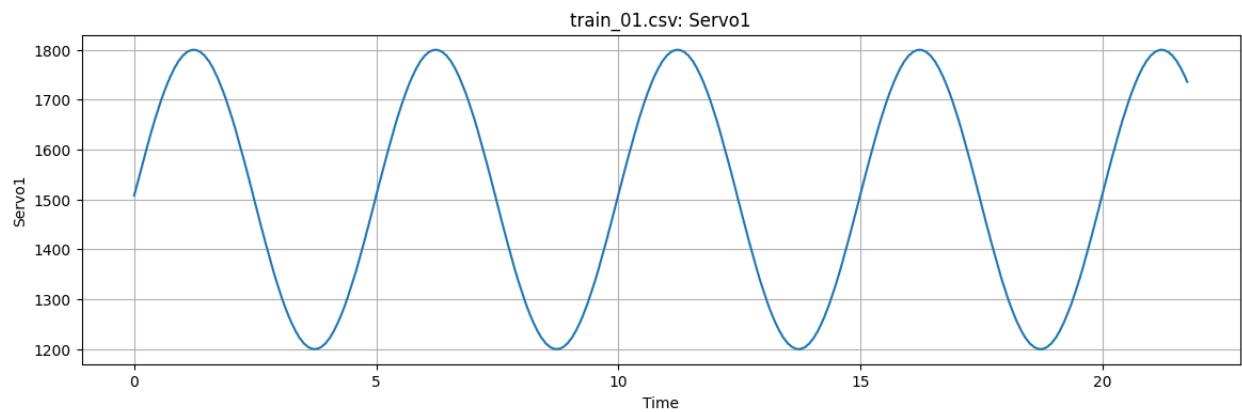
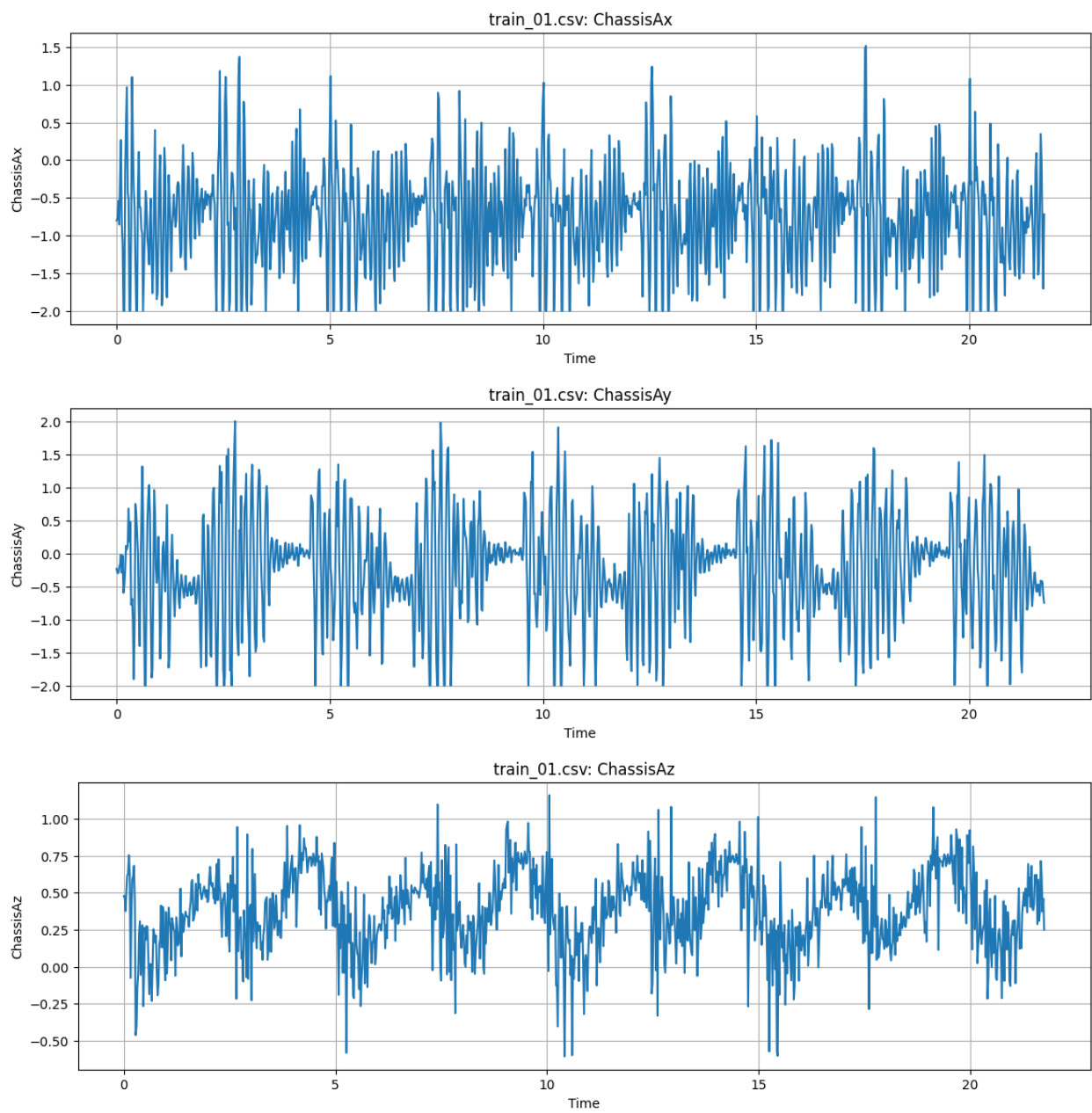


Training Data Set 1:

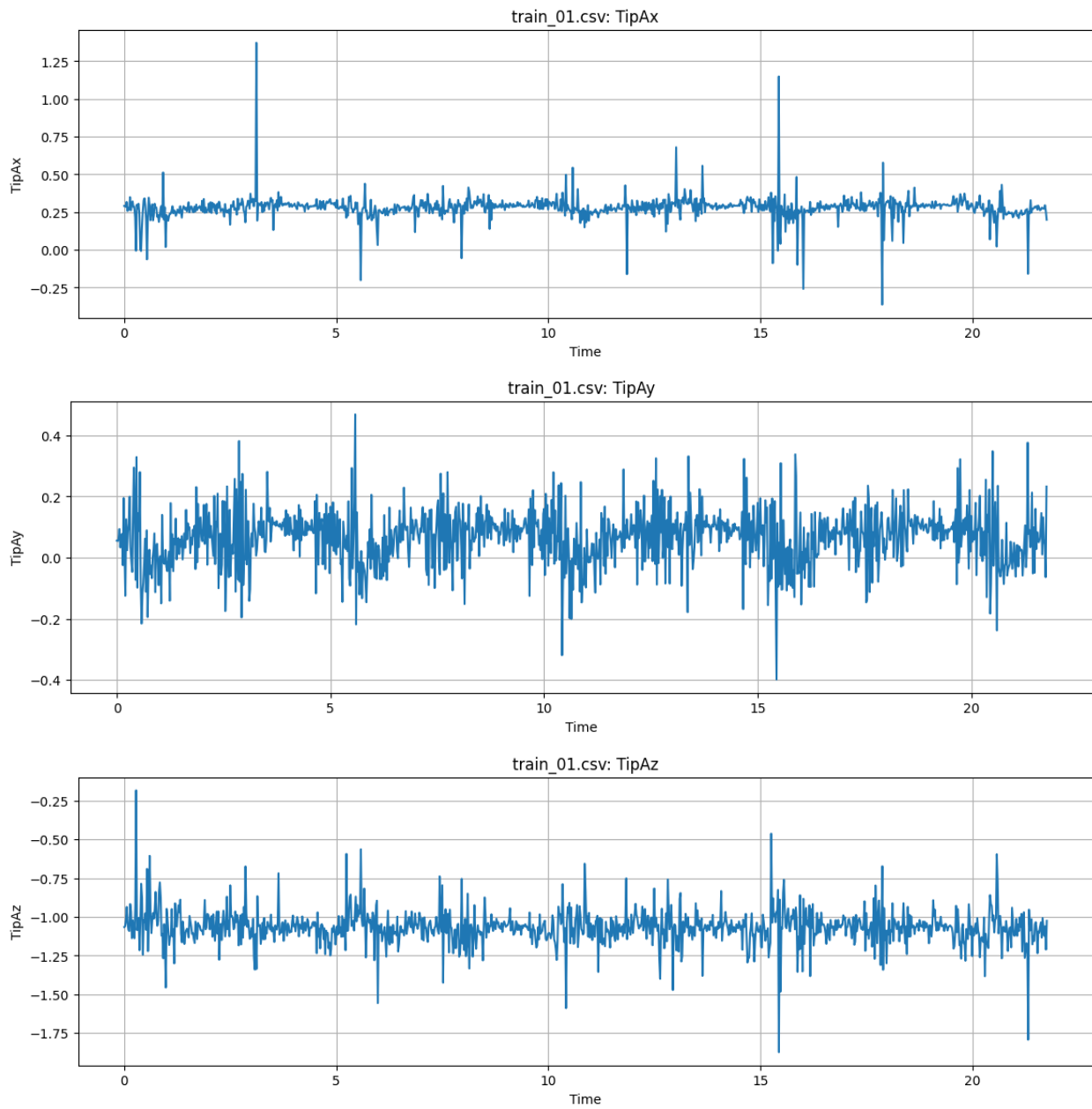
Servo Inputs:



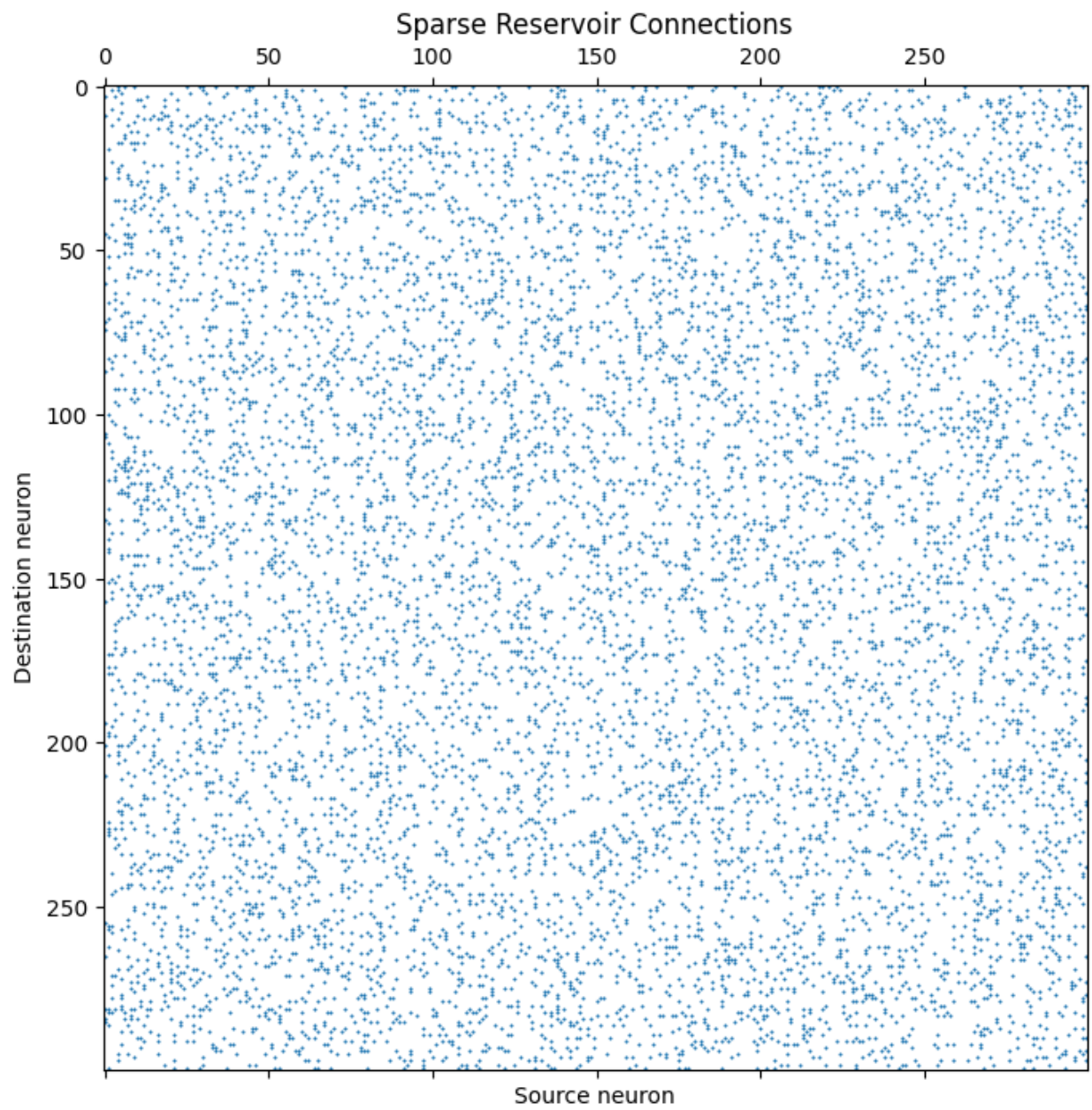
Chassis Accelerometer Axes:



Antenna tip Accelerometer Axes:

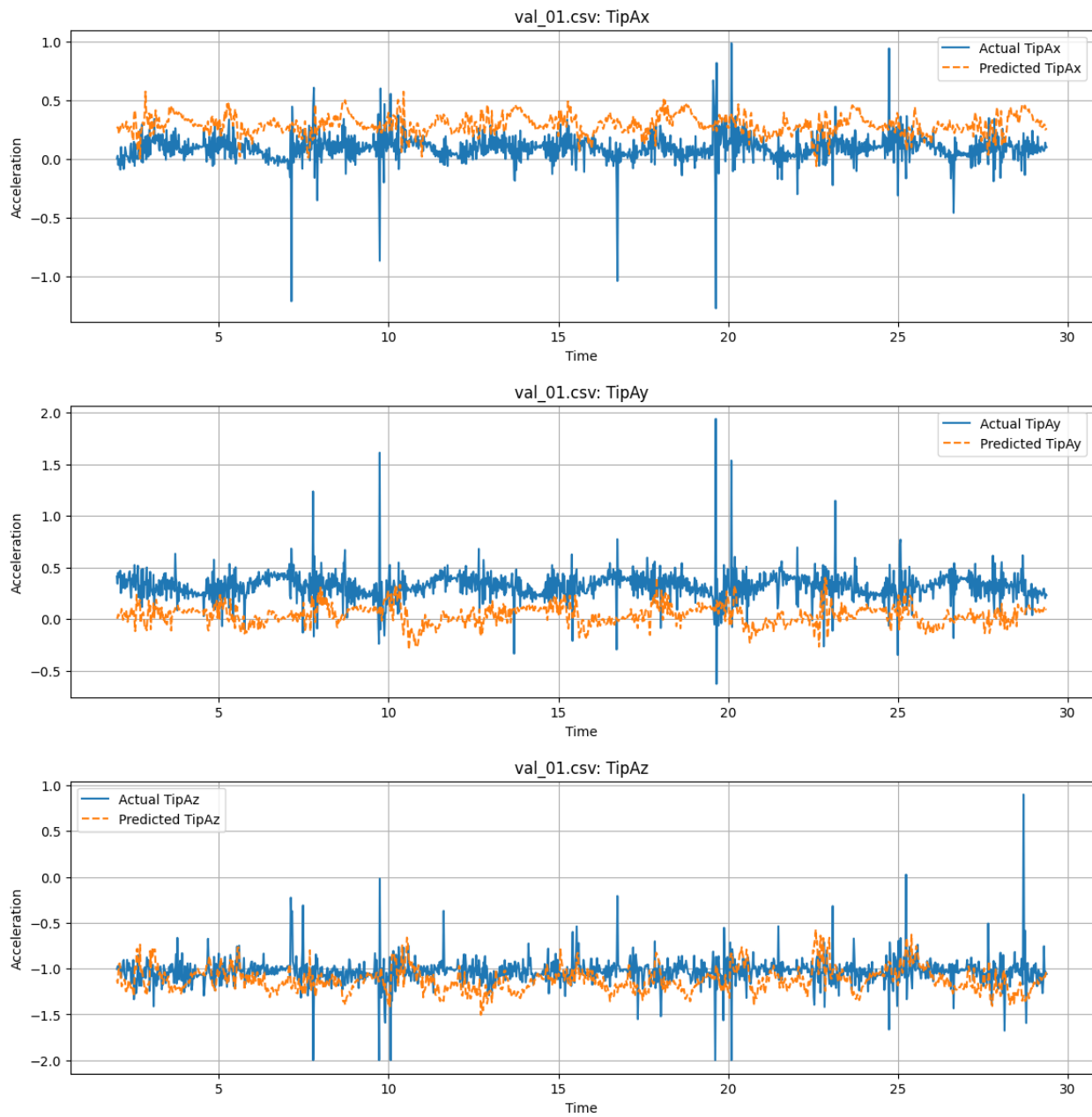


Selected sampling frequency: 50.000000000001066 Hz



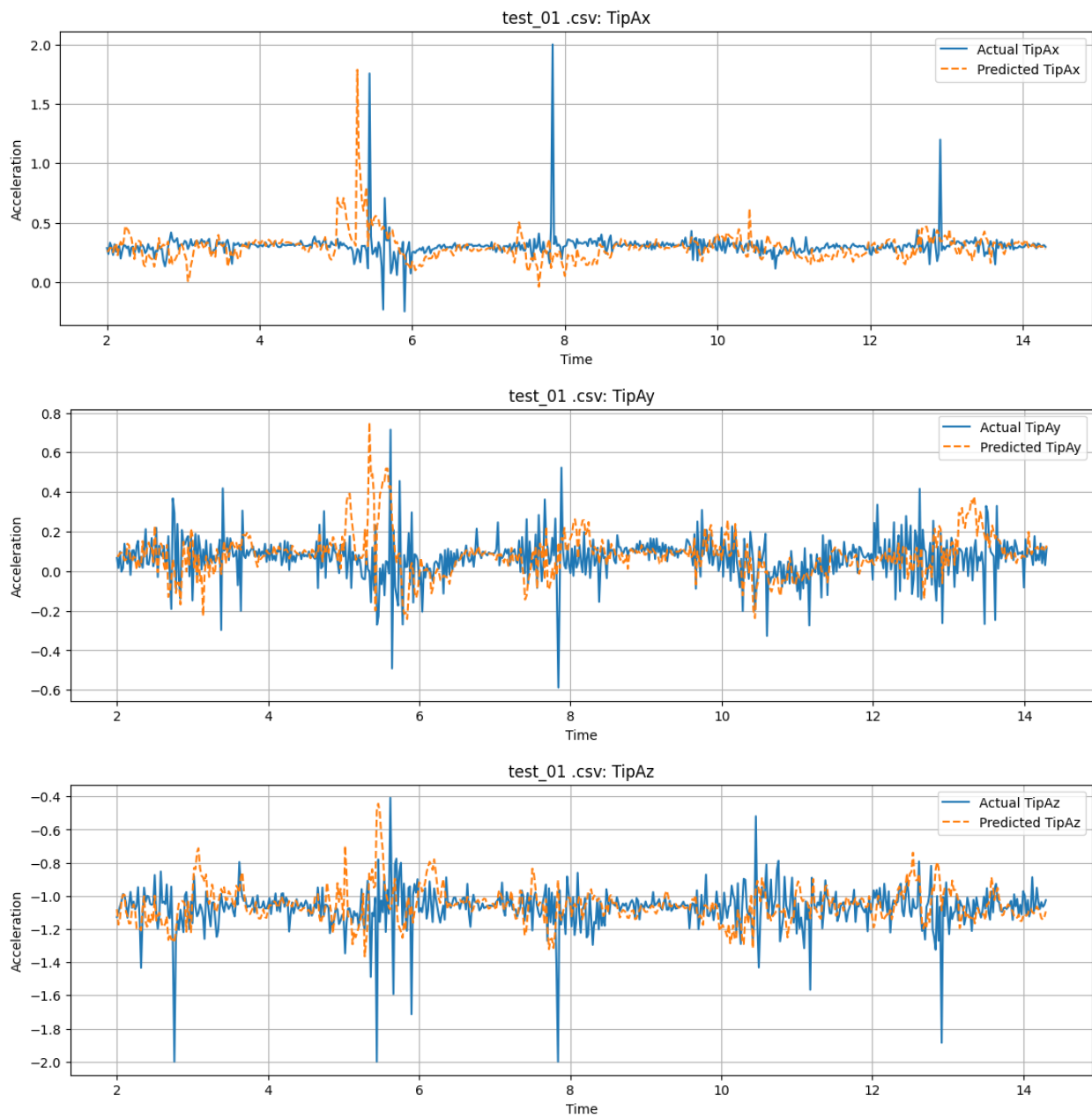
Validating Data Set 1:

Antenna tip Accelerometer Axes ESN Predicted vs Actual output:

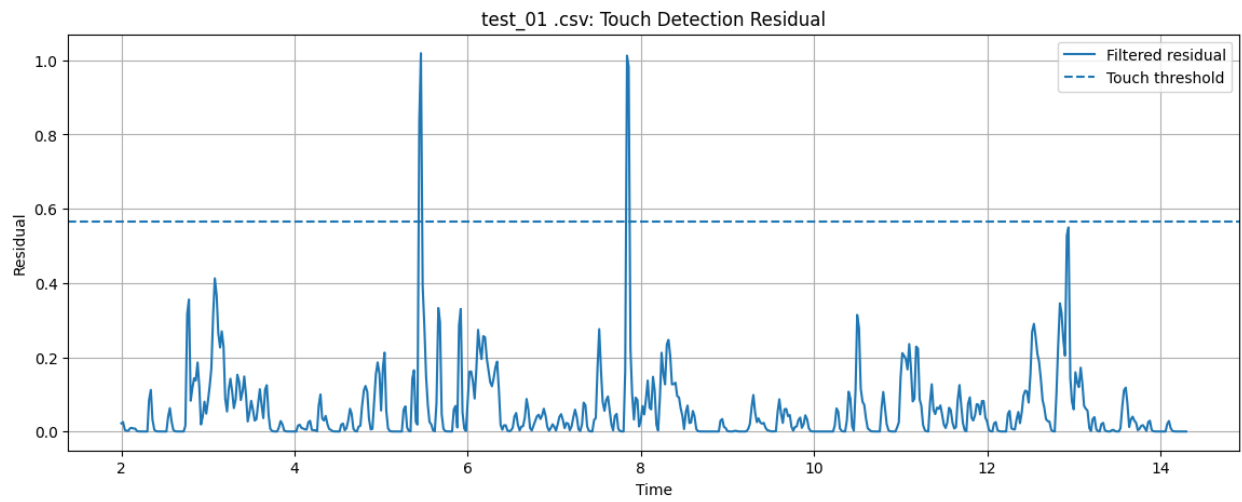


Touch (Test) Data Set 1

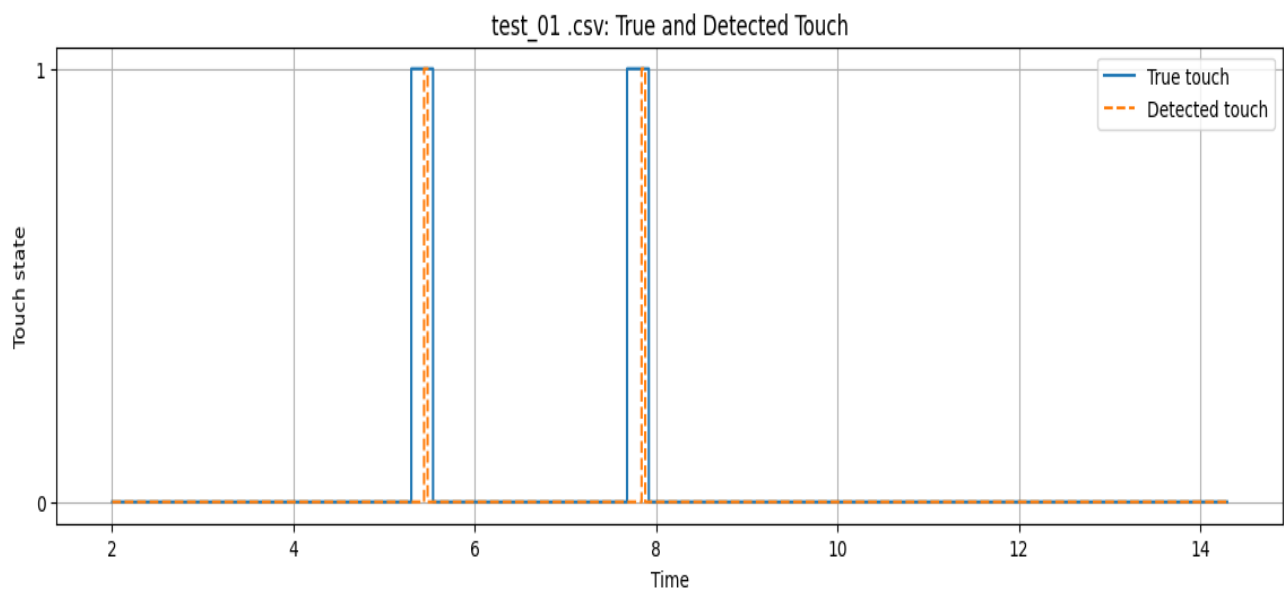
Antenna tip Accelerometer Axes ESN Predicted vs Actual output (After LPF):



Touch Detection Residual



True Touch vs Detected Touch



	precision	recall	f1-score	support
No Touch	0.9673	1.0000	0.9834	592
Touch	1.0000	0.1667	0.2857	24
accuracy			0.9675	616
macro avg	0.9837	0.5833	0.6346	616
weighted avg	0.9686	0.9675	0.9562	616

```
DetectionResult
True Negative      592
False Negative     20
True Positive       4
Name: count, dtype: int64
```